



SEMESTER 2 EXAMINATIONS 2022/2023

MODULE: CA4023 - Natural Language Technologies

PROGRAMME(S):

DS	BSc in Data Science
CSPT	PhD-track

YEAR OF STUDY: 1,4

EXAMINER(S):

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TIME ALLOWED: 3 Hours

INSTRUCTIONS: Answer Question 5 and 3 other questions. All questions carry equal marks.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.

The use of programmable or text storing calculators is expressly forbidden.

Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

There are no additional requirements for this paper.

QUESTION 1**[TOTAL MARKS: 25]****Q 1(a)****[4 Marks]**

Provide four examples of Document Classification tasks, two that can be tackled by **Logistic Regression** and two by **Multinomial Logistic Regression**.

Q 1(b)**[12 Marks]**

Briefly explain the role of the following in **Logistic Regression**:

1. Sigmoid function
2. Loss function
3. Weights
4. Regularization

Q 1(c)**[9 Marks]**

Given the following training corpus

Document	Class
<i>Too noisy</i>	negative
<i>Noisy and lacking in atmosphere</i>	negative
<i>Noisy but not too noisy</i>	positive
<i>Great atmosphere and ambience</i>	positive

what label would a **Naïve Bayes sentiment classifier** which uses the words in a document as features assign to the following “document”?

Sadly lacking in ambience

Show the steps which the classifier follows to arrive at a prediction.

[End of Question 1]

QUESTION 2

[TOTAL MARKS: 25]

Q 2(a)

[10 Marks]

Given the following training corpus

<s> x y </s>
<s> y y </s>
<s> x z </s>
<s> z y </s>

what probabilities will an **unsmoothed bigram language model** assign to the following sentences?

<s> x x </s>
<s> y </s>
<s> x y z </s>

How will these probabilities change if **add-one smoothing** is employed? Show how you arrived at your answers.

Q 2(b)

[7 Marks]

Explain the difference between **intrinsic** and **extrinsic** evaluation of language models, and state an advantage and disadvantage associated with each type.

Q 2(c)

[4 Marks]

An n-gram language model is a **count-based** language model and a neural language model is a **prediction-based** language model. Explain the difference between count-based and prediction-based language models.

Q 2(d)

[4 Marks]

GPT is an **autoregressive** language model and BERT is a **masked** language model. Explain the difference between an autoregressive language model and a masked language model.

[End of Question 2]

QUESTION 3

[TOTAL MARKS: 25]

Q 3(a)

[7 Marks]

Provide the two pieces of information that are used by a **Hidden Markov Model part-of-speech classifier**. State the additional sources of information that can be used by a **linear-chain Conditional Random Field classifier**.

Q 3(b)

[5 Marks]

Show how a Hidden Markov Model is derived from **Bayes Rule**, and explain any simplifying assumptions that are applied.

Q 3(c)

[8 Marks]

A **recurrent neural network (RNN)** can be used to perform sequence labelling tasks, e.g. part-of-speech tagging. Show, with the aid of a diagram how the following sequence

Tag me please

would be part-of-speech tagged using an RNN.

Q 3(d)

[5 Marks]

Briefly explain how an **LSTM** improves over a **vanilla RNN**.

[End of Question 3]

QUESTION 4**[TOTAL MARKS: 25]****Q 4(a)****[4 Marks]**

Explain why **word vectors** are useful in Natural Language Processing.

Q 4(b)**[8 Marks]**

Consider the following word co-occurrence matrix:

	truck	vehicle	sunlight	garage	wild
car	50	55	12	38	0
emission	25	60	2	1	1
garden	1	1	22	12	25
plant	1	1	35	2	30
flower	3	0	27	7	40

Show how the **dot product** is used to compute the similarity between the words:

1. *plant* and *flower*
2. *plant* and *car*

Show how **cosine similarity** is used to compute the similarity between the words:

1. *plant* and *flower*
2. *plant* and *car*

Q 4(c)**[4 Marks]**

Explain the difference between **sparse** and **dense** word vectors. Why are dense vectors considered to be more effective?

Q 4(d)**[4 Marks]**

What is the task learned by the **Skipgram Negative Sampling word2vec classifier**? Why are the predictions produced by this classifier typically thrown away?

Q 4(e)

[5 Marks]

Explain, using an example, the difference between **contextualised** word vectors produced by networks such as BERT and **static, non-contextualised** word vectors produced by word2vec.

[End of Question 4]

QUESTION 5

[TOTAL MARKS: 25]

Q 5(a)

[12 Marks]

Describe what you see as the three largest ethical problems with the large language model, **ChatGPT**. Explain why you have chosen these three problems over others.

Q 5(b)

[6 Marks]

For each problem identified in Q5(a), suggest a potential solution.

Q 5(c)

[7 Marks]

In your opinion, will the net effect on society of ChatGPT (and similar successor language models) be positive or negative? Justify your position.

[End of Question 5]

[END OF EXAM]